Teach & Repeat Rover Navigation

Project update | Initial simulation results

Ayush Kaulgud · Irenov Joel · Tapisar Singh

October 7, 2026

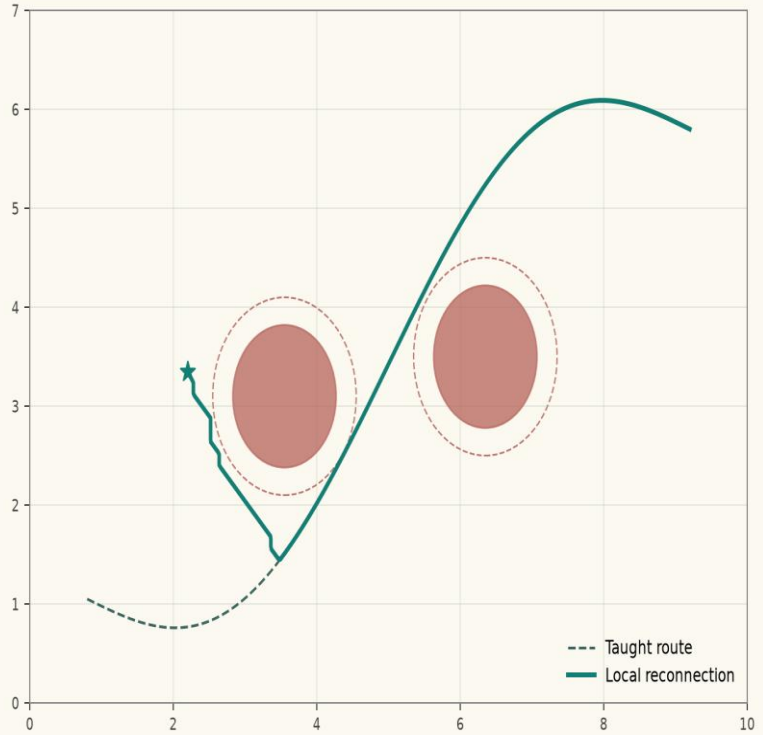
Where the project stands

The task

Start the rover away from a taught route and recover to the same goal without hitting an obstacle.

What works today

- One safe demonstration
- Four recovery planners
- Path and runtime safety checks
- A separate ICP alignment demo



A* reconnection from one displaced start. Dashed line is the taught route.

From demonstration to evaluation

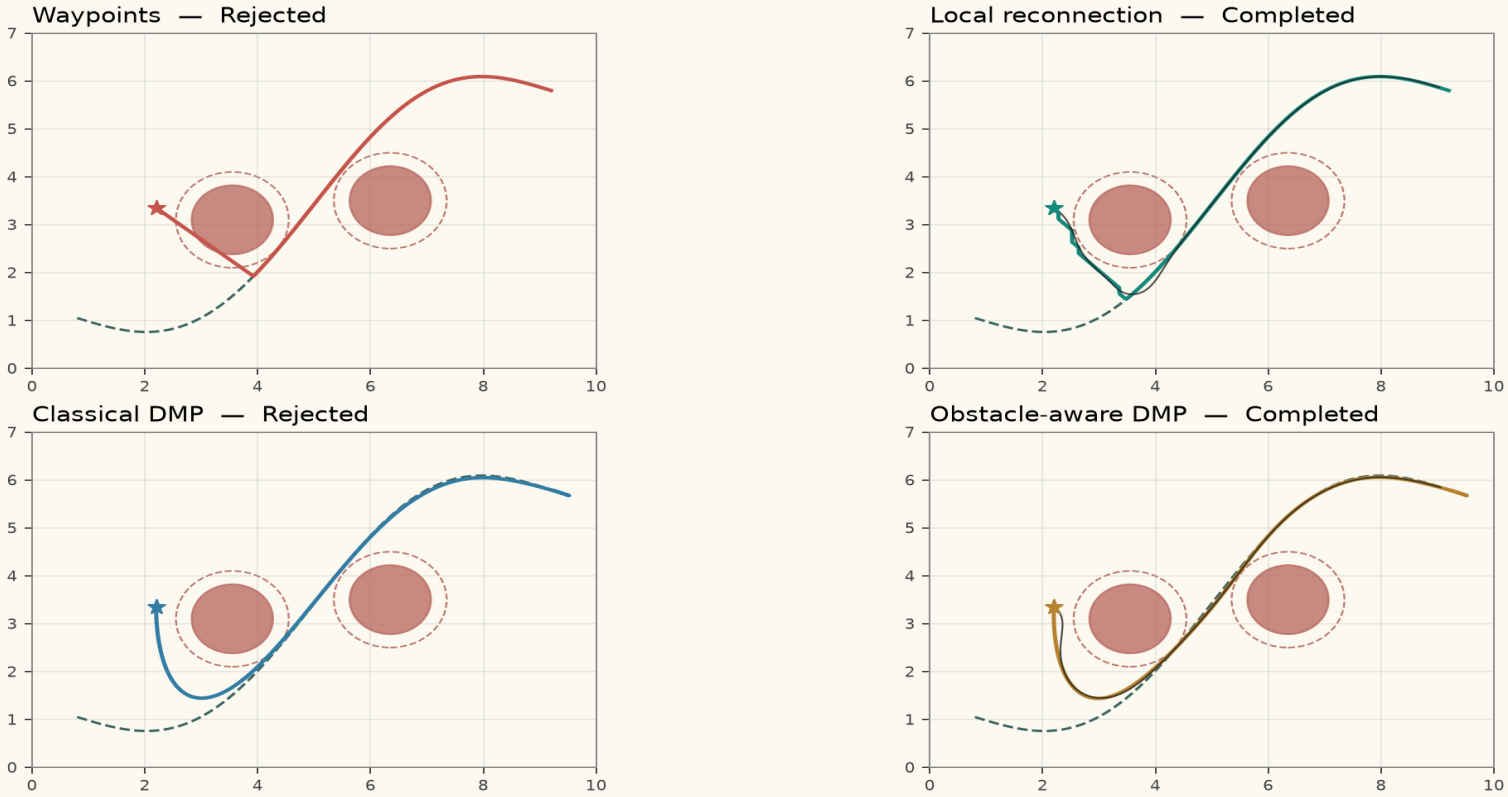


Four candidate paths

Direct waypoints · Local A* reconnection
Classical DMP · Obstacle-aware DMP

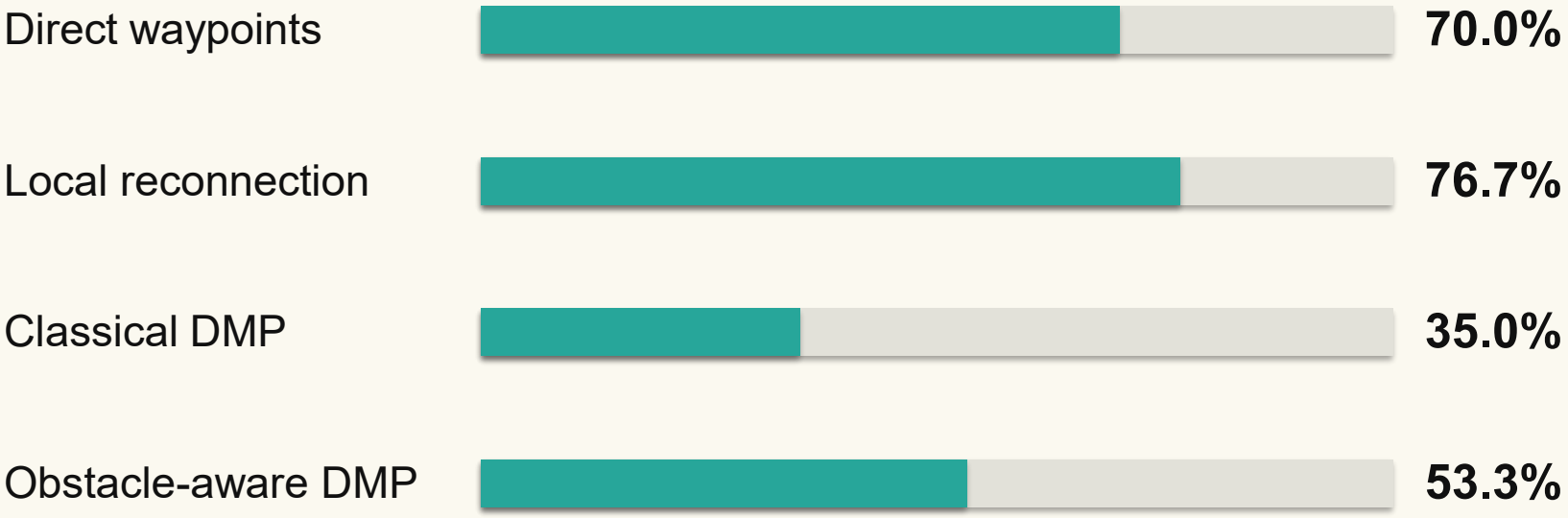
Current navigation runs use exact pose; ICP is demonstrated separately on a synthetic scan.

One offset start, four outcomes



Start: (2.2, 3.35) m, −35°. Red circles are obstacles; dotted rings include rover radius + safety margin.

Completion across 60 starts per method



A* local reconnection completed the most starts in this map.

Across all 240 trials: 0 collisions; unsafe plans were rejected or stopped.

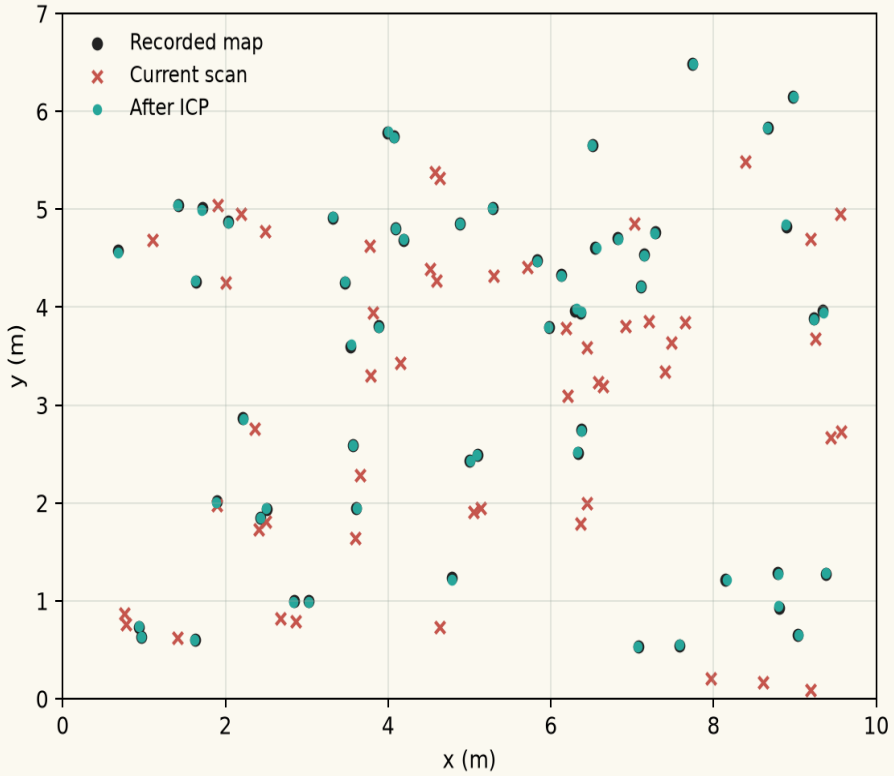
The other measurements matter too

Method	Rejected	Stopped	Min. clear.	Goal err.	Plan time
Direct waypoints	30.0%	0.0%	0.130 m	0.169 m	0.05 ms
Local reconnection	10.0%	13.3%	0.108 m	0.168 m	2.70 ms
Classical DMP	65.0%	0.0%	0.122 m	0.163 m	3.12 ms
Obstacle-aware DMP	45.0%	1.7%	0.119 m	0.165 m	4.33 ms

All methods had 0% observed collision rate. Goal error is the median of completed runs.

Minimum clearance includes sampled safety stops; planning time varies by computer.

ICP scan alignment: separate prototype



55 synthetic terrain landmarks

Alignment RMSE 0.012 m

Translation error 0.0016 m

Rotation error 0.008°

Navigation results on the previous slides still use exact rover pose.

What we can say so far

1. Local A* reconnection is strongest in this single map: 76.7% completion.

2. Obstacle coupling raises DMP completion from 35.0% to 53.3%.

3. No collisions occurred with safety checks enabled, but some paths stopped early.

Next: integrate ICP with noisy odometry; test more maps and pose offsets; tune controller tracking.

Method notes and sources

Trial design

60 starts = 4 x positions × 5 y positions × 3 headings. Static 10 × 7 m map, two circular obstacles, 0.16 m rover radius, 0.12 m margin. Shared bounded controller.

Reproduce

```
python src/robot_motion.py     python build_slides.py
```

Background references

Ijspeert et al. (2013) on DMPs; Hoffmann et al. (2009) on obstacle coupling; Besl & McKay (1992) on ICP. Sources listed in the original class presentation.